Reg. No.															

B.Tech/ M.Tech (Integrated) DEGREE EXAMINATION, NOVEMBER 2025

First Semester

21PYB102J - SEMICONDUCTOR PHYSICS AND COMPUTATIONAL METHODS

(For the candidates admitted during the academic year 2025-2026)

Note:

i. **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

ii. **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 hours **Max. Marks: 75**

PART - A (20 × 1 = 20 Marks) — Marks CO BL

Answer **all** Questions

1. Which of the following is a direct band-gap semiconductor? — 1 1 1
   (A) Ge (B) Si
   (C) GaAs (D) Both Si and Ge

2. In an indirect band-gap semiconductor, the conduction band minimum and valence band maximum are located..... — 1 1 2
   (A) At the same energy and momentum (B) At different energies but same momentum
   (C) At the same energy but different momentum (D) At different energies and different momentum

3. At a finite temperature, the probability of electron occupancy at the Fermi level E_f is: — 1 1 2
   (A) 0 (B) 0.25
   (C) 0.5 (D) 1

4. Which computational method uses the atomic orbital approach to determine electronic band structure? — 1 1 2
   (A) Cellular method (B) APW method
   (C) Pseudopotential method (D) Tight-binding method

5. As temperature increases, the Fermi level in an n-type semiconductor ..... — 1 2 2
   (A) Moves upward (B) Moves downward
   (C) Remains Constant (D) Disappears

6. The Shockley–Read–Hall (SRH) mechanism describes ..... — 1 2 1
   (A) Trap-assisted recombination (B) Direct band-to-band transition
   (C) Auger recombination (D) Photon emission only

7. A light-emitting diode (LED) emits light when it is ..... — 1 2 1
   (A) Reverse-biased (B) Not biased
   (C) Forward-biased (D) Unbiased

8. Under which bias condition does the depletion region width increase in a PN junction? — 1 2 1
   (A) Reverse bias (B) Forward bias
   (C) Both reverse and forward bias (D) No bias

9. Which of the following represents an intra band transition? — 1 3 2
   (A) Band-to-band transition (B) Impurity level to band transition
   (C) Free-carrier transition (D) Trap level to band transition

THE HELPERS

10. Fermi's Golden Rule is used to determine the 1 3 2

(A) Velocity of light (B) Momentum of electrons

(C) Absorption energy of electrons (D) Transition rate per unit volume

11. When upward transitions exceed downward transitions, what condition is achieved? 1 3 1

(A) Optical gain (B) Optical loss

(C) Equilibrium (D) Zero absorption

12. Which of the following is a major loss mechanism in photovoltaic cells? 1 3 2

(A) Production of Charge Carriers (B) Recombination of electron-hole pairs

(C) High conductivity of the material (D) Low resistivity of the material

13. The four-point probe method is superior to the two-point probe method because it 1 4 2

(A) Provides easier sample handling (B) Eliminates contact and lead resistance errors

(C) Measures pressure directly (D) Measures temperature directly

14. The hot-point probe method is primarily used to 1 4 1

(A) Thermal conductivity (B) Determine effective mass

(C) Identify semiconductor type (n or p) (D) Lattice constant

15. C–V measurements in a PN junction or MOS capacitor are used to extract 1 4 1

(A) Crystalline properties (B) Thermal conductivity

(C) Doping concentration and built-in potential (D) Lattice constant

16. The Boltzmann Transport Equation (BTE) describes 1 4 2

(A) Quantum tunneling of electrons (B) Distribution of charge carriers under external forces and scattering

(C) Diffusion of photons in a medium (D) Variation of capacitance with frequency

17. A quantum dot exhibits confinement in 1 5 1

(A) One dimension (B) Two dimensions

(C) Three dimensions (D) Four dimensions

18. Which is NOT a typical property of carbon nanotubes (CNTs)? 1 5 2

(A) High tensile strength (B) Excellent electrical conductivity

(C) Poor thermal conductivity (D) High aspect ratio

19. In Atomic Force Microscopy (AFM), surface topography is obtained by 1 5 1

(A) Measuring X-ray absorption (B) Detecting cantilever deflection of a sharp tip scanning the surface

(C) Measuring transmitted electrons (D) Using optical interference patterns

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20. TCAD (Technology Computer-Aided Design) tools are primarily used to		1	5	2
(A) Measure experimental resistivity	(B) Predict semiconductor device behavior before fabrication			
(C) Calibrate instruments	(D) Synthesize new materials			

PART - B (5 × 8 = 40 Marks) — Marks CO BL

Answer **all** Questions

21. a) Explain the postulates of classical free electron theory with its merits and drawbacks.	8	1	3
(OR)			
b) With a neat sketch explain elaborately the Fermi-Dirac distribution function and its variation with temperature.	8	1	3
22. a) With a neat diagram, explain the formation of a p-type semiconductor. Discuss the role of acceptor levels and the variation of Fermi level with temperature.	8	2	2
(OR)			
b) Explain the principle construction and working of a PN junction diode with relevant diagrams.	8	2	3
23. a) Deduce the Einstein coefficient expression for the ratio between spontaneous and stimulated emission.	8	3	3
(OR)			
b) Demonstrate the working principle of photovoltaic cell with a suitable diagram and explain the efficiency of the solar cell.	8	3	3
24. a) Define Hall Effect and derive the expression for the Hall coefficient and deduce carrier density, carrier and mobility Hall voltage.	8	4	3
(OR)			
b) Explain the principle, construction and working of hot point probe method with necessary diagram.	8	4	3
25. a) Define carbon nanotubes (CNTs) with their types. Explain its properties and its applications.	8	5	2
(OR)			
b) Explain the principle and working of scanning electron micoscopy (SEM) and explain the source and utilization.	8	5	3

PART - C (1 × 15 = 15 Marks) — Marks CO BL

Answer **any 1** Question

26. a) Derive an expression for density of states g(E) for a three-dimensional (3-D) system. Plot the graph for relationship between density of states g and energy (E).	12	1	4
b) Determine the position of the Fermi level of silicon at 300 K, if E_g = 1.12 eV, m^*_e = $0.12m_0$ and $m^*_h = 0.28m_0$	3	1	4
27. a) Explain the principle, construction and working of a Physical Vapor Deposition (PVD) Technique with their advantages.	10	5	3
b) With a neat sketch explain the working of linear four probe method.	5	5	3

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